

UK Visa Application Guidance & Document Readiness Agent

Engineering & Product Justification

1. Product Rationale

The project was chosen around a practical problem: people applying for UK visas often know what they want to achieve but do not necessarily know where to find the relevant official information.

GOV.UK already provides the information, so the purpose of the agent is not to replace the official source. Instead, the agent helps users find and understand relevant information more efficiently.

The product is therefore positioned as a **guidance and document-readiness assistant**.

The primary user is a prospective UK visa applicant, although the same system could potentially support immigration consultants and caseworkers as a research assistant.

2. Why an Agent Rather Than a Basic Chatbot?

A simple LLM chatbot could answer questions, but it would not provide enough control for this use case.

The agent needs to decide whether it should search the knowledge base, use an eligibility tool, use previous conversation context or respond that the required information is unavailable.

For example:

"I have a sponsor, passport and English qualification. Am I eligible and what documents do I need?"

This question involves several operations. The agent needs to understand the request, plan the steps, use the appropriate tools, retrieve supporting information and then combine the results.

This makes an agent architecture more appropriate than a single LLM API call.

3. Architecture

The implemented workflow can be summarised as:

User → Planning → Tool Selection → RAG / Eligibility Tool → Gemini → Memory / Feedback → Final Response

The RAG component uses official GOV.UK guidance as the primary knowledge source.

Documents are loaded and split into smaller chunks. The chunks are converted into embeddings and stored in ChromaDB. When a question is received, semantic retrieval identifies the most relevant chunks. These are then supplied to the LLM as context.

This reduces the need for the LLM to answer purely from its general knowledge.

4. Technology Choices

Python was used as the main programming language because it is well suited to rapid AI prototyping and has strong libraries for LLM, retrieval and data-processing workflows.

Google Gemini was used as the LLM because it provides the generation and tool-calling capabilities needed by the project.

LangChain was used to organise the document-processing, retrieval and agent workflow. It provides useful components for document loading, text splitting, vector stores and LLM integration.

ChromaDB was selected as the vector store because it is lightweight and suitable for a prototype running in Google Colab.

Google Colab provided a convenient development environment without requiring local infrastructure.

5. Why RAG?

RAG is particularly important for this project because visa guidance is a knowledge-intensive domain and official requirements can change.

A model answering purely from its training knowledge may provide an answer that sounds convincing but is not supported by the project's current guidance.

With RAG, the workflow is:

User question → Semantic search → Relevant GOV.UK chunks → LLM context → Answer

This provides a stronger connection between the final answer and the source material used by the project.

However, RAG is not a guarantee of correctness. If the document is missing, outdated or not retrieved correctly, the response can still be incomplete.

6. Tool Usage and Planning

Two tools were introduced to demonstrate agent-based behaviour.

The first is the document search tool, which allows the agent to retrieve relevant information from the vector database.

The second is a basic eligibility-checking tool.

The tools are deliberately limited and controlled. This is preferable to allowing the model to perform unrestricted operations.

Planning was then introduced so that the agent can decide when multiple steps are necessary.

For example:

Eligibility question → Eligibility tool → Document search → Final response

This demonstrates a more realistic agent workflow.

7. Memory and Adaptive Behaviour

Short-term memory was introduced to improve the quality of multi-turn conversations.

For example:

"What documents do I need for a Skilled Worker visa?"

followed by:

"What about my wife?"

The agent can use the earlier context to interpret the second question.

Adaptive behaviour was introduced separately. User feedback such as:

"Please answer using short bullet points."

can be stored and used to modify later responses.

Memory retention is intentionally limited. The system should retain useful conversation context but should not unnecessarily retain sensitive personal information.

8. Safety and Guardrails

Safety was treated as an important engineering requirement because incorrect immigration guidance can have serious consequences.

The agent therefore has several boundaries:

- Use official guidance as the primary RAG source.
- Do not invent information when retrieval fails.
- Restrict the intended use case to UK visa guidance.
- Use controlled tools.

- Handle errors without crashing.
- Make the limitations of the system clear.
- Do not present the system as a replacement for professional immigration advice.

These safeguards reduce risk but do not eliminate it.

9. Engineering Trade-Offs

The project deliberately uses a relatively simple architecture because the goal was to demonstrate the capabilities of an AI agent within a capstone project.

ChromaDB is appropriate for the prototype but would not necessarily be the final choice for a large production system.

Similarly, the eligibility tool is deliberately limited. Attempting to encode every part of the UK Immigration Rules would make the project considerably more complex and would introduce additional maintenance and accuracy risks.

The current design therefore prioritises **clarity, testability and reproducibility**.

10. Product Success and Future Development

The product should ultimately be measured by whether it reduces the effort required to find relevant official information while maintaining reliable responses.

Useful measures include:

- Retrieval accuracy
- Answer accuracy
- Hallucination rate
- Tool-selection accuracy
- Consistency
- Response time
- User satisfaction

The most important future development would be automatic monitoring of GOV.UK guidance. When guidance changes, the relevant documents should be downloaded again, processed and re-embedded.

Other future improvements include secure document uploads, better eligibility rules, stronger retrieval evaluation, authentication, production monitoring and larger-scale safety testing.

11. Conclusion

The engineering decisions were made to solve the specific problem rather than simply demonstrate an LLM.

The combination of **RAG, tool usage, planning, memory, adaptive behaviour and error handling** gives the agent capabilities that a basic chatbot would not have.

At the same time, the project deliberately recognises its limitations. The prototype is best used to help users find and understand official UK visa guidance, rather than making immigration decisions on their behalf.